

# Luca Tomei

Software Engineer · IT Development Advisor

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## Professional Summary

Software Engineer (MSc Engineering in Computer Science, Sapienza University of Rome) building enterprise systems since 2021: full-stack development with .NET 8, Angular and SQL Server, Infor M3 ERP integration (on-premise and cloud) through ION, MEC and ESB, microservices, REST APIs and low-code delivery on OutSystems.

IT Development Advisor at Yokohama TWS, hands-on in a 4-person IT team: development standards, CI/CD and containerisation strategy, about 15 integration flows with 3PL partners, technical oversight of outsourced work. End-to-end ownership from requirements to deployment and support. Cybersecurity enthusiast: firmware reverse engineering, NFC/RFID research on Flipper Zero, hardened self-hosted infrastructure.

## Technical Skills

**Languages:** C#, Python, TypeScript, JavaScript, Java, SQL, PHP, C/C++, Scala, Swift, Kotlin, PowerShell

**Backend & APIs:** .NET 8, Laravel, Node.js, REST APIs, Microservices, RabbitMQ, XSD/XML, JSON

**Frontend & Mobile:** Angular, Vue.js, React, React Native, SwiftUI

**ERP & Integration:** Infor M3 (on-premise and cloud), Infor ION, MEC, ESB, 3PL integrations, OutSystems

**Databases & Data:** SQL Server, MySQL, PostgreSQL, MongoDB, SQLite, Databricks

**DevOps & Cloud:** Azure DevOps, GitHub Actions, CI/CD, Docker, Linux, IIS, Proxmox, Cloudflare, Azure

**Embedded & Security:** Firmware reverse engineering, NFC/RFID (MIFARE Classic), Flipper Zero, ESP32, MQTT, Home Assistant, Node-RED, network and VPN configuration

**Design & CAD:** Autodesk Fusion 360, AutoCAD, 3D printing

**Ways of working:** Agile, code review, requirements analysis, technical documentation, cross-functional collaboration, problem solving, mentoring

## Work Experience

### IT Development Advisor

February 2026 – Present

*Yokohama TWS (formerly Trelleborg Wheel Systems) – Tivoli (Rome), Italy*

- Define development strategy and best practices: microservices, Azure DevOps CI/CD pipelines, Docker-based deployments
- Develop hands-on within a 4-person IT team (one manager, three advisors), owning code quality and performance on .NET 8, Angular and SQL Server
- Design and govern enterprise API orchestration between microservices and legacy systems via Infor ION, MEC and ESB
- Coordinate the Infor M3 cloud migration and the rollout of OutSystems low-code applications, two of them already in production
- Own about 15 integration flows with 3PL logistics partners and the technical oversight of outsourced developments; turn business requirements into documented technical specifications

### IT Development Specialist

May 2023 – February 2026

- Led technical analysis, development and implementation of custom solutions on Infor M3 ERP (on-premise) and OutSystems
- Developed REST APIs and C#/.NET backend microservices integrated with Angular dashboards and SQL Server databases
- Designed and maintained about 15 enterprise integration flows (MEC, ESB) with 3PL partners: XSD/XML schemas, data mapping, troubleshooting of inbound/outbound documents
- Built Azure DevOps CI/CD pipelines replacing manual multi-step releases with automated, repeatable deployments of frontend, backend and database
- Performed code reviews and root cause analysis; maintained technical documentation and runbooks

*Stack:* .NET 8, C#, Angular, TypeScript, SQL Server, MySQL, PHP/Laravel, Vue.js, RabbitMQ, Infor M3, Infor ION, MEC, ESB, OutSystems, Azure DevOps, Docker, IIS

### ERP Developer

November 2021 – May 2023

*Trelleborg Wheel Systems – Tivoli (Rome), Italy*

- Managed the technical side of the Infor M3 ERP (Business Engine, interfaces, layouts, reports) and its integration with other business systems
- Analysed functional documentation, wrote technical documentation, fixed bugs and kept ERP versions aligned with vendor releases
- Supported the business on technical ERP issues and worked with vendor consultants on complex implementations

### Teaching Assistant, Functional and Parallel Programming

October 2021 – December 2021

*Sapienza University of Rome – Rome, Italy*

- Tutored the online section (about 100 students) of the Scala-based course held by Prof. Camil Demetrescu: labs, programming assignments, quizzes and exams

### Software Developer Intern – OBDA Systems, Rome (web front end: React, REST)

July 2018 – September 2018

### Home Automation and IT Technician (freelance)

November 2015 – October 2021

*Self-employed – Rome, Italy*

- Installed home automation and control systems for homes and shops; built databases for business data management; provided technical consulting

## Education

### Master's Degree in Engineering in Computer Science – 110/110

January 2020 – October 2021

Sapienza University of Rome – Rome, Italy

- Thesis: *Design and proof-of-concept of the integration of an electric scooter with a wheelchair: from the mechanical design to the firmware upgrade* – custom firmware with adaptive acceleration, Python client/GUI for firmware builds, 3D-designed bracket

### Bachelor's Degree in Computer and Automation Engineering – 99/110

2016 – 2019

Sapienza University of Rome – Rome, Italy

- Thesis: *Mapping Management of an Ontology-based Data Access System* (Java, React)

### Scientific High School Diploma – Liceo Scientifico Ettore Majorana, Guidonia (Rome)

2009 – 2014

## Projects

### Electric Scooter Wheelchair – [github.com/LucaTomei/Electric-Scooter-Wheelchair](https://github.com/LucaTomei/Electric-Scooter-Wheelchair)

2020 – 2021

- Built a working prototype coupling an electric scooter to a wheelchair: 3D-printed CAD bracket and reverse-engineered firmware with acceleration and braking tuned for the new load (MSc thesis) (Python, firmware, Fusion 360, 3D printing)

### OPBG Telegram Booking Bot

Private project

- Built an invite-only multi-user bot automating the Bambino Gesù children's hospital booking portal: availability monitoring with notification or auto-booking, separate bot and worker processes, idempotent booking protocol, admin panel (Python, aiogram, SQLAlchemy, MySQL, Docker)

### Flipper Mi Band NFC Writer – [github.com/LucaTomei/flipper-miband-nfc-writer](https://github.com/LucaTomei/flipper-miband-nfc-writer)

2025

- Developed a Flipper Zero app that reads, writes, emulates and verifies NFC data on Xiaomi Mi Band devices: automatic backups, sector-level progress, hex diff viewer, persistent settings and exportable logs (C, Flipper Zero, NFC, MIFARE Classic)

### Telegram Audio Transcriber – [github.com/LucaTomei/Telegram-Audio-Transcriber](https://github.com/LucaTomei/Telegram-Audio-Transcriber)

2026

- Built a self-hosted bot transcribing voice messages and audio locally with Whisper models; optional text cleanup via OpenAI-compatible endpoints, whitelist and admin roles, no audio persistence, Docker Compose deployment from CI-built images (Python, Faster-Whisper, ffmpeg, SQLAlchemy, Docker, GitHub Actions)

### LazioHealthMonitorBot – [github.com/LucaTomei/LazioHealthMonitorBot](https://github.com/LucaTomei/LazioHealthMonitorBot)

2025 – Present

- Created a bot used by about 35 people to poll the Lazio regional health service every five minutes: real-time notifications, filters, direct or automatic booking of the first free slot, PDF confirmations, CI and container images on every release (Python, Telegram Bot API, REST, Docker, GitHub Actions)

### Carburanti Bot – [github.com/LucaTomei/Carburanti-Bot](https://github.com/LucaTomei/Carburanti-Bot)

2026

- Developed a real-time fuel price bot for Italy with about 12 active users: favourite station, nearby and best-price search from a shared location, daily notifications, keyboard UI and admin panel (Python, Docker, Portainer, MIMIT/Osservaprezzi open data)

### Estrai Allegati – [github.com/LucaTomei/estrai-allegati](https://github.com/LucaTomei/estrai-allegati)

2026

- Developed a native macOS app extracting embedded files from Office documents: dependency-free ZIP, OLE2/Compound File and EMF parsers written from scratch, universal binary, automated DMG releases (Swift, SwiftUI, GitHub Actions)

### Computer Scientists Retrieval – [github.com/LucaTomei/Computer\\_Scientists](https://github.com/LucaTomei/Computer_Scientists)

2020

- Ranked computer scientists by influence with PageRank and HITS over a Wikipedia/DBpedia graph, with category classification (Python, NetworkX, SPARQL/DBpedia, BeautifulSoup)

### Other open source

2019 – Present

- MAXG30CFW – Ninebot Max G30 firmware builder (Python, reverse engineering) · telegram-ha-monitor – multi-server monitoring bot with leader election and RBAC · Esp32-Cam-PIR-Telegram – motion-triggered camera controlled via Telegram (C++) · Hoodie (iOS, Swift) · ColliGO – queue-management Telegram bot for shops during COVID-19 (freelance, 2020)

## Certifications

OutSystems Associate Reactive Developer – OutSystems, 2024

Academy Accreditation – Databricks Fundamentals (2025); Databricks Lakehouse Fundamentals (2022) – Databricks

Software Architecture Fundamentals; Artificial Intelligence Fundamentals: Machine Learning; Cloud Computing: Core Concepts – Birdify – LMSAFP, 2024

Security: Introduction to Cybersecurity Tools & Cyber Attacks (IBM, Coursera, 2022); Cybersecurity Foundations, Cybersecurity Threat Landscape, Cloud Security Awareness, DevOps Foundations: DevSecOps (LinkedIn Learning, 2024)

Introducing Semantic Kernel: Building AI-Based Apps; Agile Software Development; APIs and Web Services; Test Automation;

Behavior-Driven Development; Project Management Foundations; Data Engineering; Generative AI and Prompt Engineering – LinkedIn Learning, 2024

Introduction to iOS Mobile Application Development – Meta, Coursera, 2023

Foundations of Project Management; Foundations of User Experience (UX) Design – Google, Coursera, 2022

Introduction to Cloud Computing; Introduction to Data Analytics – IBM, Coursera, 2022

Crash Course on Python; Using Python to Interact with the Operating System – Google, Coursera

REST API (Intermediate); SQL (Basic, Intermediate); Java, Python, React (Basic) – HackerRank, 2021

## Languages

Italian – Native · English – B2 (CEFR) · French – A2 (CEFR)

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